



DeepLearning.AI

AI Prompting for Everyone

Finding Information



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AI Prompting for Everyone

The AI novice and
the AI power user

Asking hard questions

AI novice

Uses AI for simple questions

Does Taco Bell still have the Double Decker Taco?



Not on the permanent menu, but it often returns for limited runs



This is cool but not that life-changing



AI power user

Gives AI difficult tasks and time to think

Car specs, quotes, insurance plans



What are the tradeoffs for each car? Read everything and think hard before answering



Here's a detailed report...



VS



This saved me hours!



Providing the right context

AI novice

Uses short prompts
& hopes AI will fill in the blanks

Write a good self-review to
send to my boss



Dear boss,

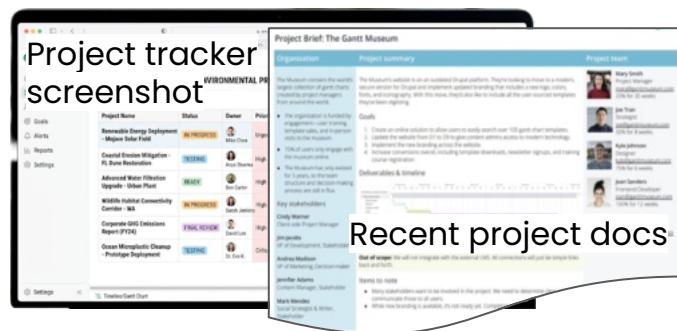
I'm not just a good employee. I'm an
employee that matters. Let me
delve into why...

*This doesn't reflect my
work at all...*



AI power user

Adds quality context,
files, and images



Voice
notes



Recent project docs

Write a self-review to send to
my boss. Here's what I did...



*This really captures what
I'm most proud of!*



Getting honest feedback

AI novice

Asks biased questions and gets biased answers

I have a great business idea: mobile tie-dying. Critique it?



Great idea! That's a novel and creative business.



Are you just telling me what I want to hear?



AI power user

Asks neutral questions & uses rubrics for critique

	Dimension	Excellent (16-20 Points)	Adequate (10-15 Points)	Poor (0-9 Points)
	1. THE PROBLEM & MARKET			
	2. SOLUTION & VALUE PROP			
	3. COMPETITIVE ADVANTAGE			
	4. BUSINESS MODEL			
	5. FEASIBILITY & EXECUTION			

Analyze the following business idea objectively: mobile tie-dying. Use the rubric.



8 / 100. Here's why it scores low...

Oh, well. I'm glad I have a clear perspective!



Researching

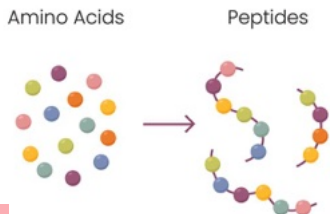
AI novice

Relies on chatbot knowledge & web search tools

Pros and cons of taking peptides as a supplement?



Peptides are short chains of amino acids. Effects depend heavily on how you take it...



I already knew that...



AI power user

Uses deep research

Pros and cons of taking peptides as a supplement?



*I'll find high quality medical sources...
Searching Mayo Clinic...
Searching Google Scholar...*



Here's your report with citations.

These results give me the full picture.



Writing

AI novice

Asks AI to just write stuff

Write a blog post about the Blackberry



In the ever-evolving landscape of mobile technology, few devices have left as profound a mark as the BlackBerry phone....



This sounds like slop.



AI power user

Starts from outline, then critiques, iterates, and expands



notes.txt

Outline a blog post about the Blackberry based on my notes



Here's an outline...

Skip section 3. Add in an anecdote.



Here's the revised outline...

Great. Expand into bullet points.



... and more back and forth!

I'm feeling inspired to finish writing!

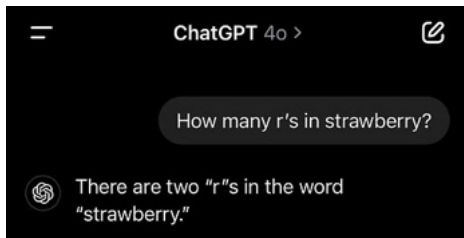
Andrew Ng



AI is more than its high profile mistakes

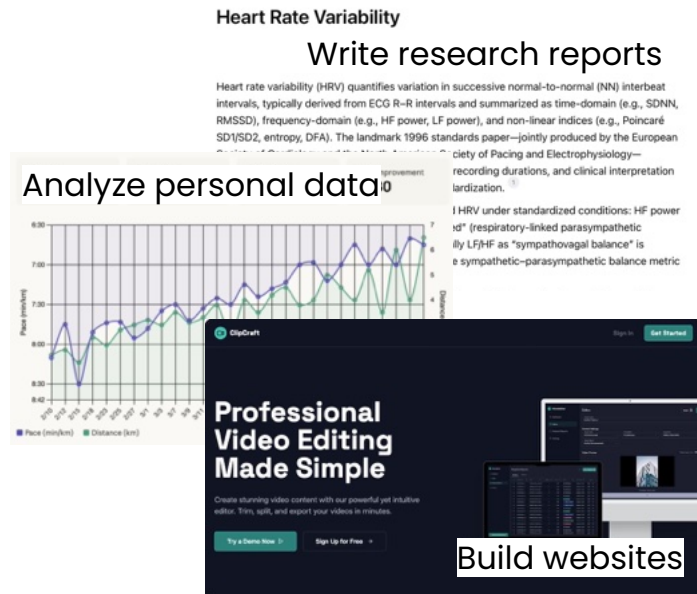
AI limitations are important, but AI power users can work around these limitations

Silly mistakes



Not representative
of AI capabilities

Real value



Being an AI power user will benefit you



Save time &
improve your life



Build cool things



Develop an
in-demand skill

These videos:

**Wherever
you are!**

to

AI power user





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Finding Information

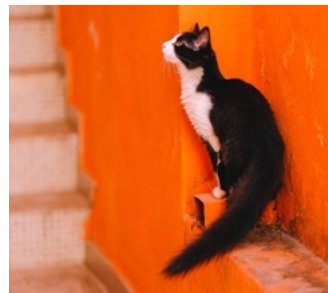
Pretrained knowledge

AI models can answer questions on a variety of topics

I dropped my phone in soup. What should I do?



Oops! Soup is more damaging than water because it contains salt and fat. Here are the **steps you can take...**



Why do cats stare at walls like they're seeing ghosts?



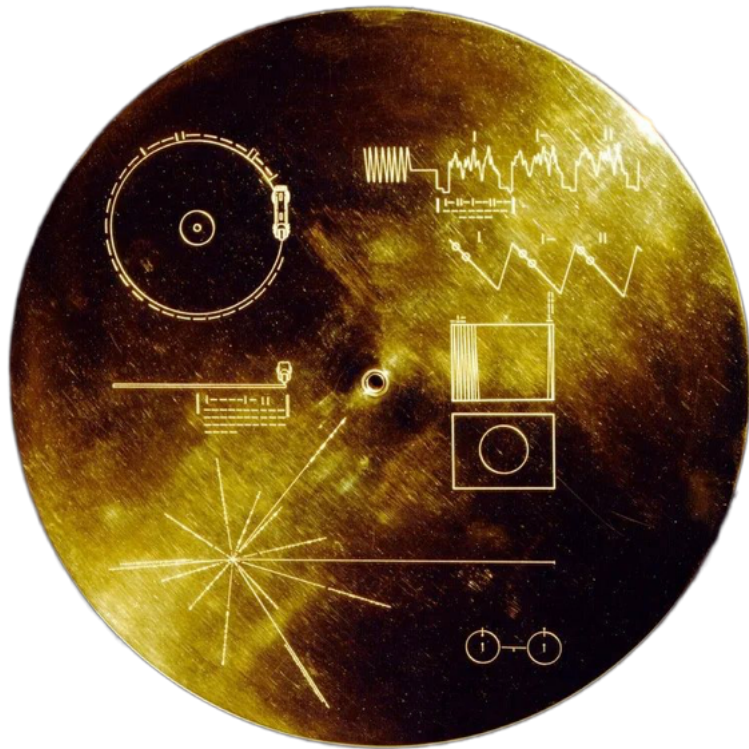
Cats can detect **subtle sounds and movements**—like your neighbor's footsteps—that humans usually miss.

They even possess niche knowledge

What kind of things were on the vinyl record sent into space?



The Voyager 1 Golden Record contains Beethoven, frog croaks, greetings in 55 languages, and more!



AI models are trained on many sources

Training on these sources produces the **pretrained knowledge**

Don't worry too much about this term!



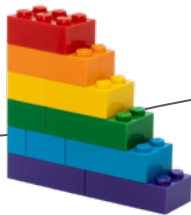
What are your
"MUST watch" films?

Reddit



LEGO Micro Cities

Jeff Friesen



Fairy bread

Wikipedia



How a Shoe Invented by a Dancer
Might Be a Game Changer

New York Times



Body Proportions and Gliding
Adaptations of Flying Squirrels

Richard Thorington, Jr., Lawrence Heaney



AI models are trained on many sources

Training on these sources produces the **pretrained knowledge**



Internet forums



Quora



Books



Encyclopedias



WIKIPEDIA



WIKIDATA



News sites

The New York Times

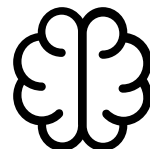
BBC



Academic articles

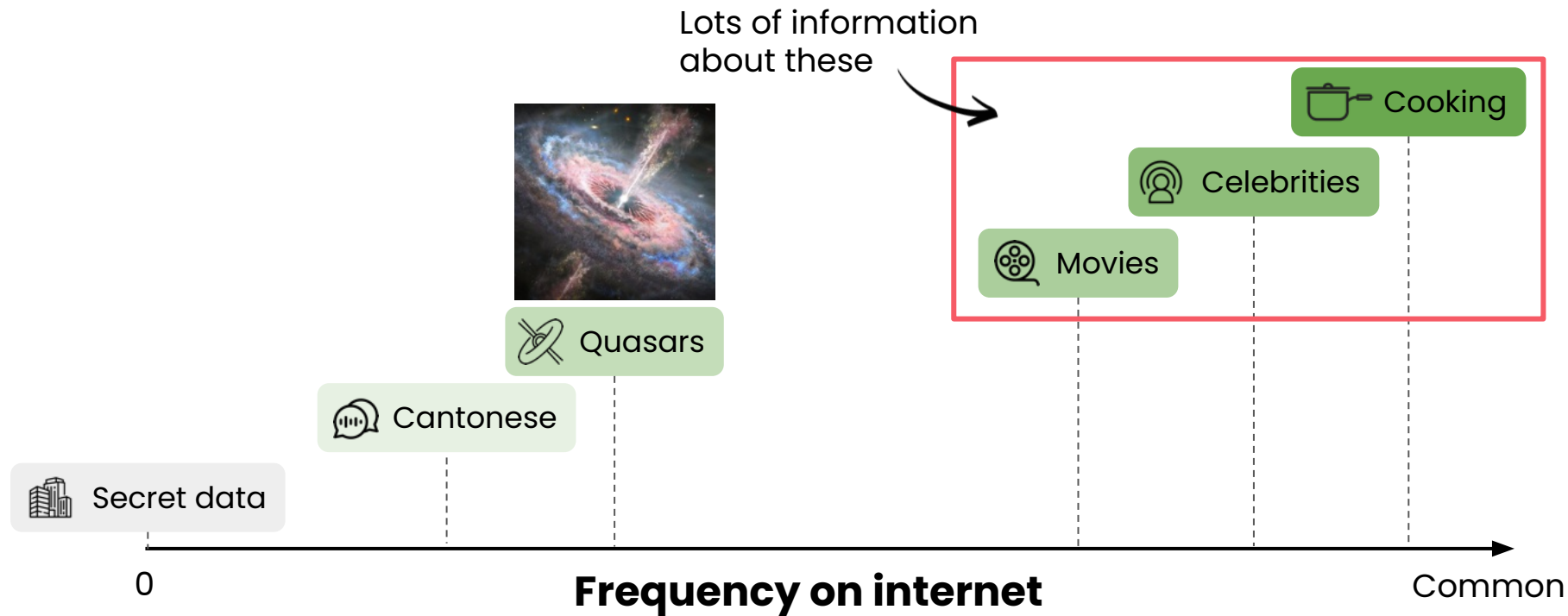
arXiv

PubMed



AI model's
brain

Pretrained knowledge reflects patterns in training data



AI models can exhibit surprising understanding

can u cook eggs in microwav



Can you cook eggs in the microwave?



Yes, but only with certain methods.

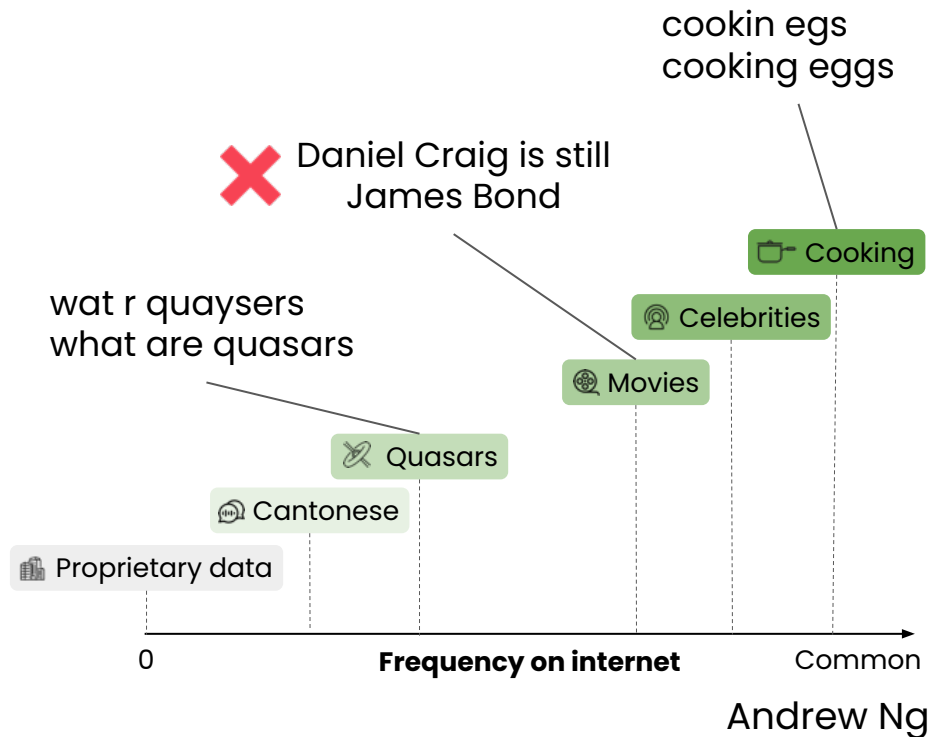
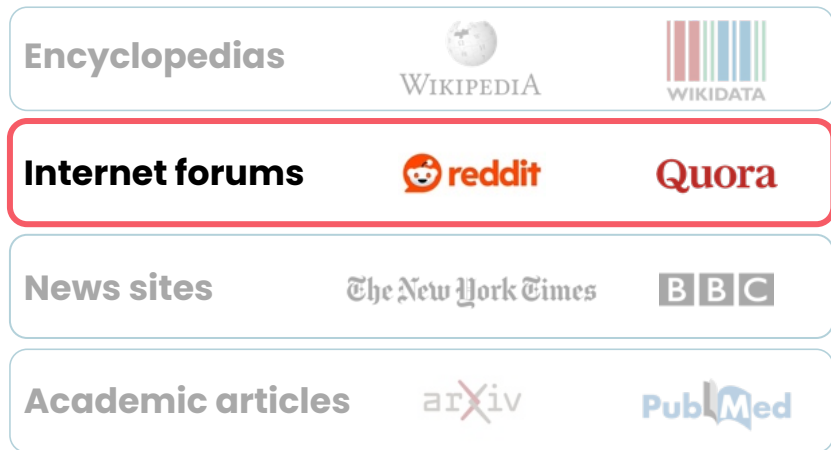
Here are three ways to **avoid egg-splotions** when cooking eggs in the microwave...



AI models are trained on public data with errors

Sources can include:

- Typos
- Misconceptions
- Outdated information





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Finding Information

Web search

Some questions trigger web search

May benefit from updated info!

What is the 6 7 meme from 2025?



Searching...

MENTAL FLOSS



Forbes

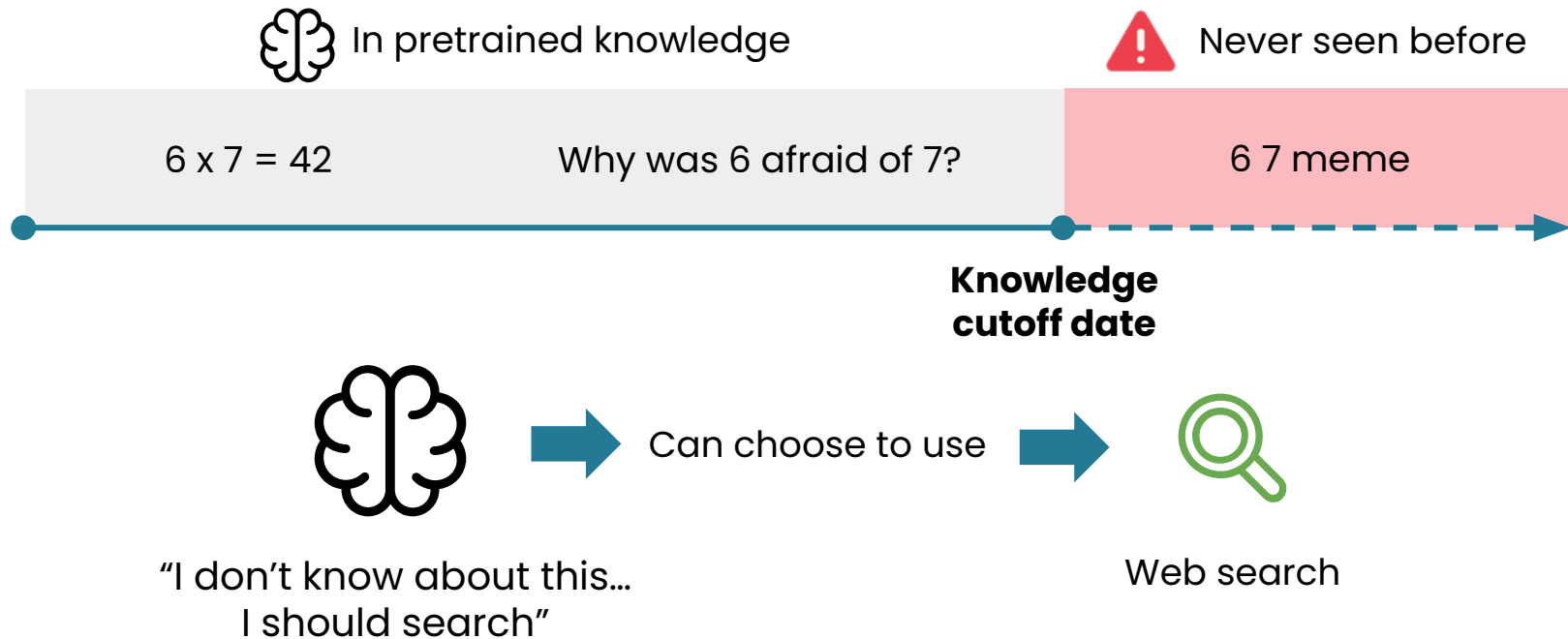


The **"6-7" meme** is a viral internet slang trend widely shared on TikTok, Instagram Reels, and YouTube Shorts.

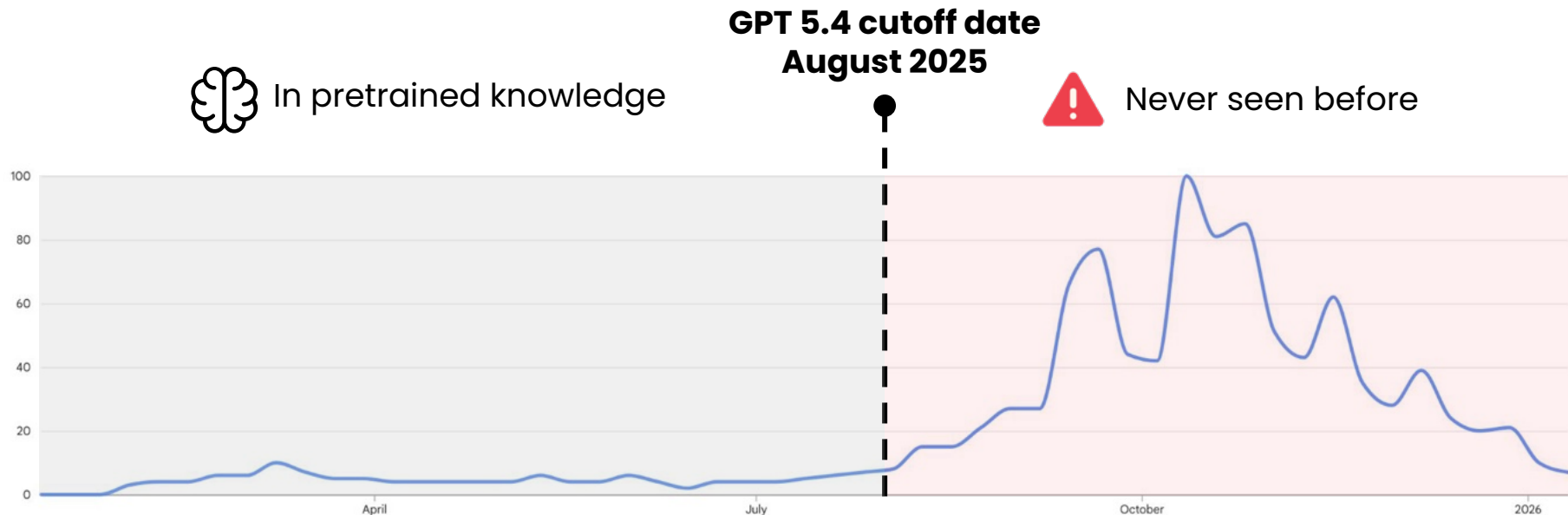


Pretrained knowledge is frozen in time

But, the internet **continues to evolve** over time



Google searches for “What does 6-7 mean?”



Different types of questions trigger search



Find me a highly rated gym
near Mountain View, CA



Searching...



Here is a list of the best **gyms in
your area...**



What is the Marquette Mountain
Cheese Roll?



Searching...



This is a popular event where
participants chase a wheel of cheese
down a hill. It happens in Michigan.

When AI models need web search

Question	Source	Reason
Help, I dropped my phone in soup.		Common knowledge on the Internet
Why do cats stare at walls?		
What was on the Voyager 1 record?		
What is the 6 7 meme from 2025?		Real-time
Find me a highly rated gym near Mountain View, CA		Location-specific
What is the Marquette Mountain Cheese Roll?		Niche

Web search can be triggered in two ways

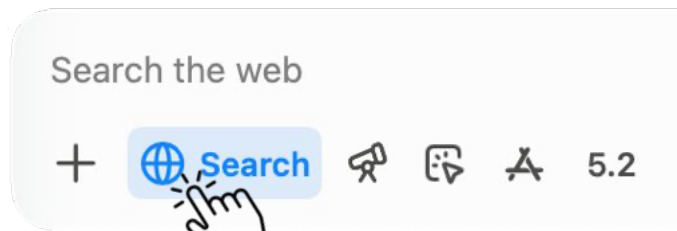
AI model decides

AI model



Is user asking for
real-time, location-specific,
or **niche** information?

You ask for it



Search the web...





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Finding Information

Web search sources

Encourage trusted sources

How safe are gray market peptides?



Searching...



honest peptide

To tell if a product is "safe", look past the marketing and verify the raw lab data...



How safe are gray market peptides? Only use sources from **official health organizations**. Look at studies **backed by rigorous science**.



Searching...



World Health Organization



EMA
EUROPEAN MEDICINES AGENCY

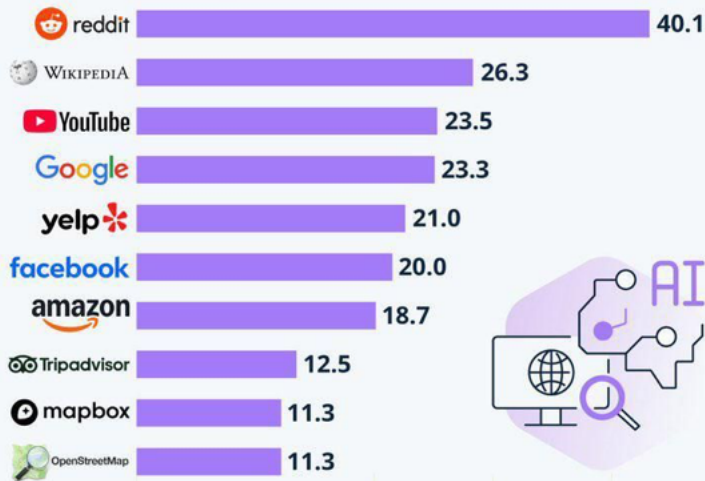
This is what official institutions say about gray market peptides and other supplements...



Web search draws from **popular** sources

Where AI Gets Its Info: Top Sources 2025

Top 10 web domains cited by large language models (LLMs) in June 2025* (in %)



* Google's AI Mode, AI overviews, ChatGPT and Perplexity
Based on 150 thousand citations from 5,000 randomly selected keywords from Semrush database.
Source: Semrush



statista



Without steering, models pull from what's most **available**—not most reliable.

Reliable sources





Web search draws from **popular** sources

How safe are gray market peptides?



VS

How safe are gray market peptides? Only use sources from **official health organizations**. Look at studies **backed by rigorous science**.



**Social media,
blogs, forums**



**Reliable
sources**





Web search can draw from **outdated** sources

Help me find places to run in Henderson, Nevada.



Here is a list of places where you can go for a jog in Henderson, NV...

Running in Henderson, NV

Other Running Related Topic

23 years ago 04/03/2003 12:23am EST

Reply New Thread Top Threads Moderation Information Q

David Small ▾

registered

Follow ▾

Block ▾

#1

23 years ago 04/03/2003 12:23am EST

Anyone know of any good places to run in Henderson, NV. I'll be there for a few days next week.

Thanks

0 0



Web search is a multi-step process

Your
customer
service team!

User-facing
AI model



Assistant
AI model



You send
a prompt



Topic
Task
Target sources
Desired depth

Web search
based
on prompt

Scan results

Filter out
irrelevant
results

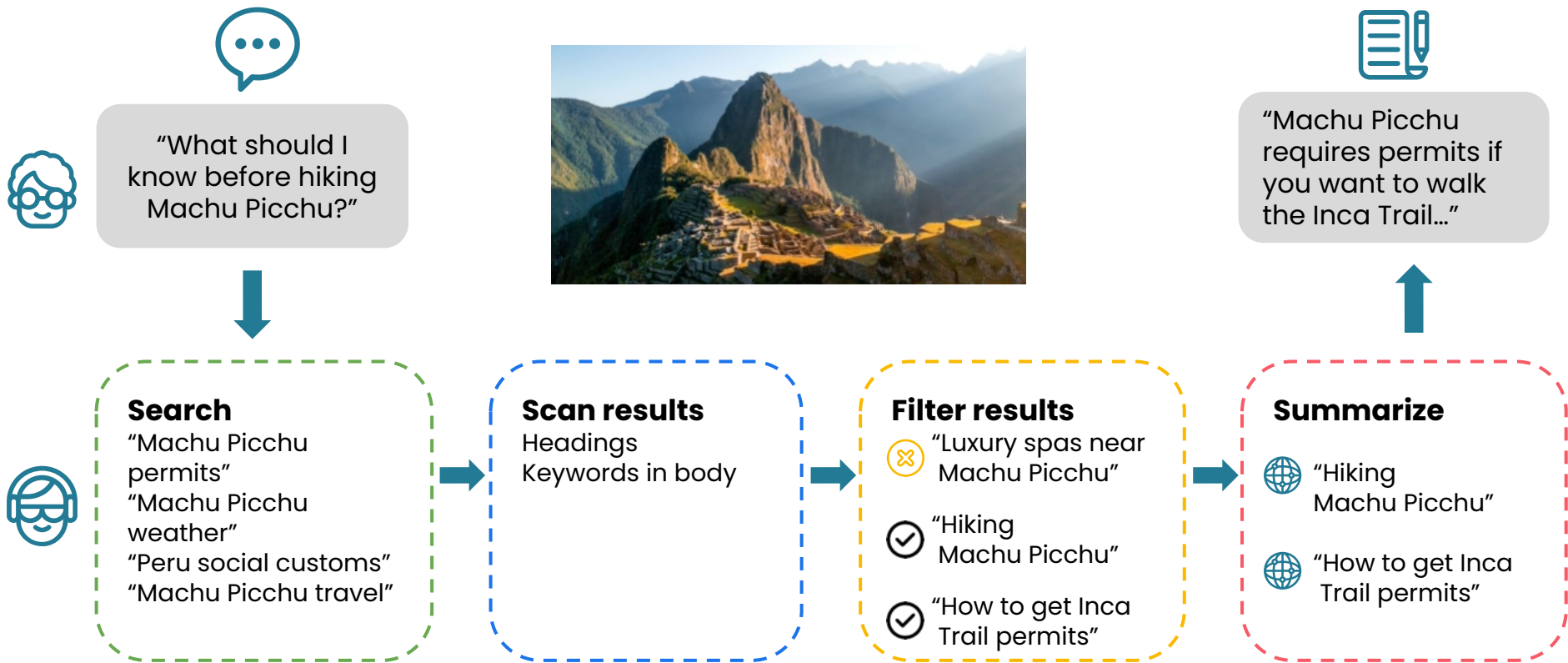
Summarize
findings

Generate
answer

Not read all pages in
their **entirety!**



Web search is a multi-step process



Choosing between AI & a search engine

Search engine

- Quickly scan several sources
- Navigate to a website
- Need data in original form

2013 Honda Civic air filter



The Home Depot

<https://www.homedepot.com> › ... › Auto Parts

Air Filter - Auto Parts



Walmart

<https://www.walmart.com> › ... › Honda Civic Air Filters

Honda Civic Air Filters

AI model

- Need a synthesis
- Information is complex
- Contrast multiple sources

Pros & cons of vitamin supplements?



Searching...



This is what **official institutions** say...



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Finding Information

Using deep research

Your Halloween haunted house

Help me set up a haunted house in my front yard for Halloween:

- Palo Alto, California
- Front yard is ~500 sq feet
- Full experience: fog machines, sound effects, actors, and a steady flow of trick-or-treaters



Here is a research plan:

1. Palo Alto & California laws
2. Decoration ideas

...

Go ahead!



Your Halloween haunted house

-  Gathering Palo Alto's rules on permits, Halloween ordinances, and safe practices for haunted houses
-  Read [ca.gov](https://www.ca.gov), [paloalto.gov](https://www.paloalto.gov)...
-  Piecing together noise levels, permits, and fire safety guidelines for Halloween events
-  Read [wikipedia.org/wiki/Fire_safety](https://www.wikipedia.org/wiki/Fire_safety) ...
-  Looking for Halloween decoration ideas
-  Read [pinterest.com](https://www.pinterest.com), [spirithalloween.com](https://www.spirithalloween.com)...
-  Writing report

Agentic AI: model makes decisions by itself on what to do next



Your Halloween haunted house



Your Halloween Haunted House



Create ▾

1. Structural and Regulatory Framework
...
2. Fire and Electrical Safety
...
3. Scare Actors and Crowd Management
...
4. Decoration Plan
...



Your Halloween haunted house



Your Halloween Haunted House



Create ▾

1. Structural and

...

2. Fire and Elect

...

3. Scare Actors

...

4. Decoration Pl

...

 Web page

 Infographic

 Quiz

 Flashcards

 Audio Overview

[Describe your own app](#)





Palo Alto Regulatory Framework

Before a single decoration goes up, we must ensure strict adherence to local municipal codes. Palo Alto enforces specific ordinances regarding noise control and public right-of-way access.

Municipal Code Checklist

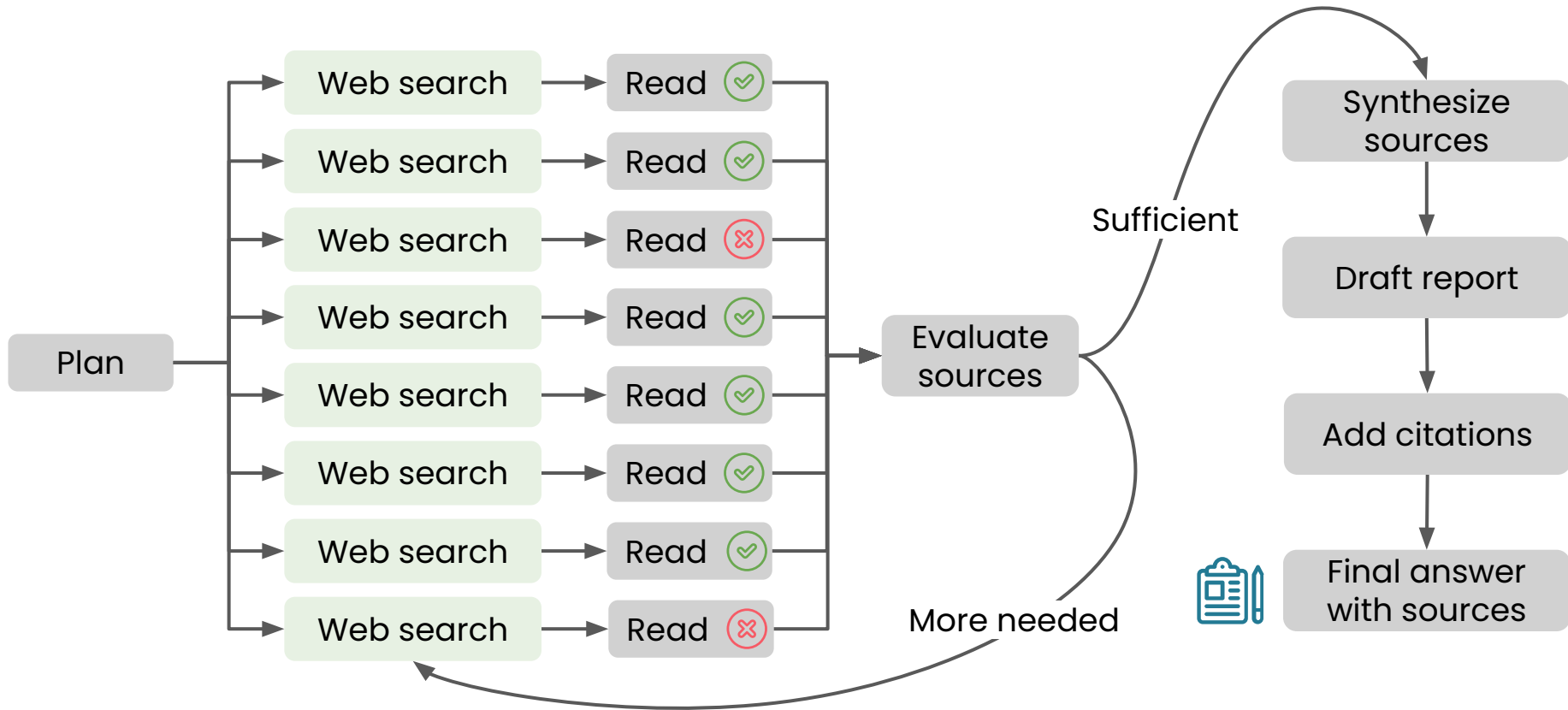
Select items as you verify them to increase your Readiness Score.

- ☐ Walkway Width > 36 inches (ADA)
- ☐ Cords taped down / covered
- ☐ Noise source directed away from street
- ☐ Fire Extinguisher accessible
- ☐ Notify immediate neighbors

Noise Ordinance Visualization

Palo Alto noise limits drop significantly after 10:00 PM. Plan your audio loop accordingly.

Example deep research flow



Web search vs. deep research

Web search



1. Find me a highly rated gym nearby
2. What's the weather in Dubai this week?

 **Questions to answer:** single

 **Human answer time:** seconds

 **Sources:** a handful

 **Trigger:** automatically or by you

Deep research



1. Impact of daily steps on long-term health?
2. How does weather affect tourism in Dubai?

 **Questions to answer:** multiple

 **Human answer time:** minutes to hours

 **Sources:** dozens to hundreds

 **Trigger:** usually by you

Finding information: Recap

	Pretrained knowledge 	Web search 	Deep research 
Example	Help, I dropped my phone in soup	Find me a highly rated gym nearby	Impact of daily steps on long-term health?
# of sources	None	Handful	Dozens or more
Freshness	Not relevant	Up-to-date	Up-to-date
AI time spent	A few seconds	Many seconds	Many minutes
Most useful for	Facts, definitions, summaries	Real-time, location-specific, or niche info	Complex synthesis